

Exceptional Control Flow: Signals and Nonlocal Jumps

Introduction to Computer Systems

Instructors:

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ECF Exists at All Levels of a System

■ Exceptions

- Hardware and operating system kernel software

■ Process Context Switch

- Hardware timer and kernel software

■ Signals

- Kernel software and application software

■ Nonlocal jumps

- Application code

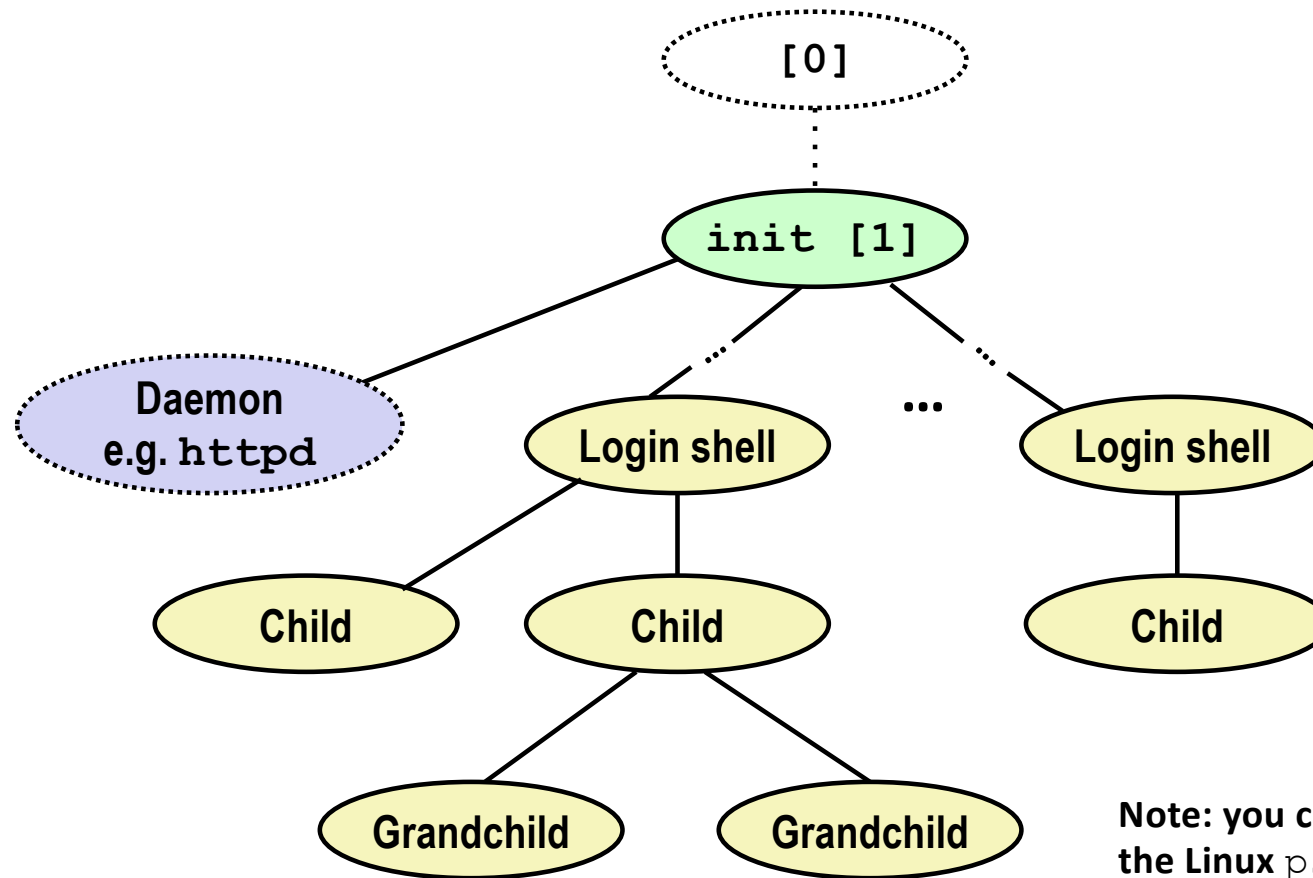
} **Previous Lecture**

} **This Lecture**

Today

- Shells
- Signals
- Nonlocal jumps

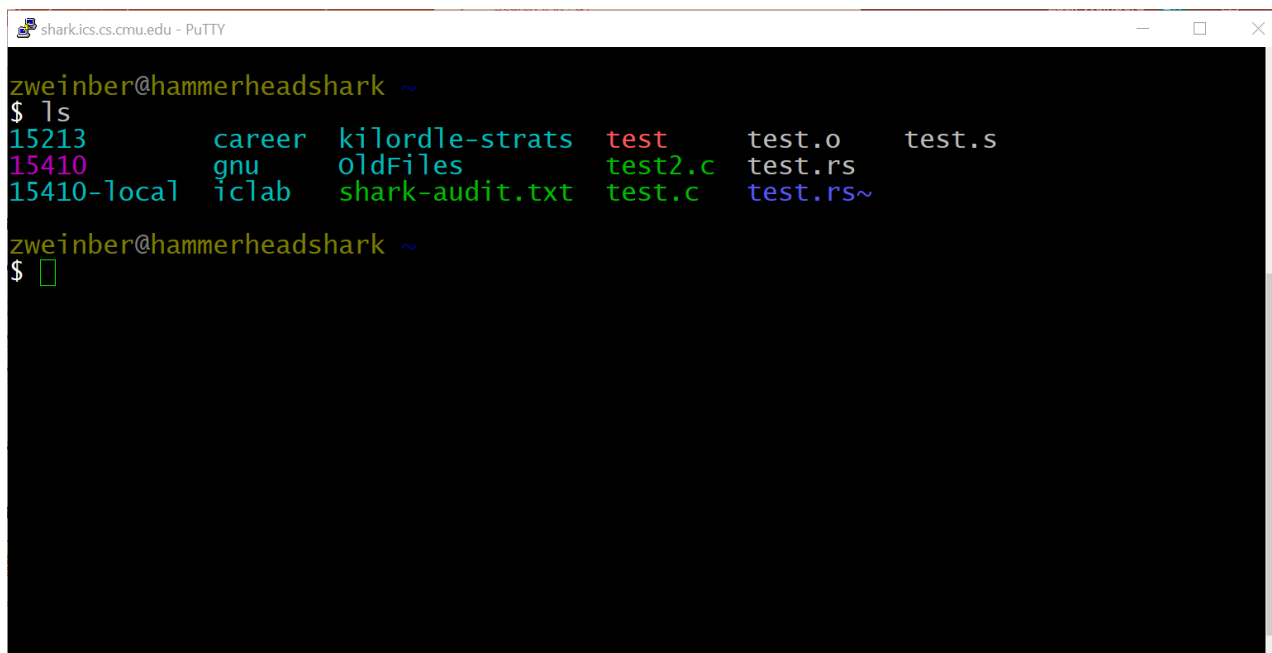
Linux Process Hierarchy



Note: you can view the hierarchy using the Linux `ps tree` command

Shell Programs

- A **shell** is an application program that runs programs on behalf of the user.
 - **sh** Original Unix shell (Stephen Bourne, AT&T Bell Labs, 1977)
 - **cshtcsh** BSD Unix C shell
 - **bash** “Bourne-Again” Shell (default Linux shell)



A screenshot of a terminal window titled "shark.ics.cs.cmu.edu - PuTTY". The prompt is "zweinber@hammerheadshark ~". The user has entered the command "\$ ls", and the output is a long listing of files and directories. The output is as follows:

```
zweinber@hammerheadshark ~  
$ ls  
15213      career  kilordle-strats  test      test.o  test.s  
15410     gnu     OldFiles        test2.c   test.rs  
15410-local iclab   shark-audit.txt  test.c    test.rs~
```

The prompt is now "\$ " with a cursor.

Shell Programs

■ Simple shell

- Described in the textbook, starting at p. 753
- Implementation of a very elementary shell
- Purpose
 - Understand what happens when you type commands
 - Understand use and operation of process control operations

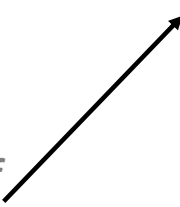
```
linux> ./shellex
> /bin/ls -l csapp.c
-rw-r--r-- 1 bryant users 23053 Jun 15 2015 csapp.c
> /bin/ps
  PID TTY          TIME CMD
 31542 pts/2        00:00:01 tcsh
 32017 pts/2        00:00:00 shellex
 32019 pts/2        00:00:00 ps
> /bin/sleep 10 &
32031 /bin/sleep 10 &
> /bin/ps
  PID TTY          TIME CMD
 31542 pts/2        00:00:01 tcsh
 32024 pts/2        00:00:00 emacs
 32030 pts/2        00:00:00 shellex
 32031 pts/2        00:00:00 sleep
 32033 pts/2        00:00:00 ps
> quit
```

Simple Shell Implementation

■ Basic loop

- Read line from command line
- Execute the requested operation
 - Built-in command (only one implemented is `quit`)
 - Load and execute program from file

*Execution is a sequence of
read/evaluate steps*



```
int main()
{
    char cmdline[MAXLINE]; /* command line */

    while (1) {
        /* read */
        printf("> ");
        fgets(cmdline, MAXLINE, stdin);
        if (feof(stdin))
            exit(0);

        /* evaluate */
        eval(cmdline);
    }
}
```

shellex.c

Simple Shell eval Function

```
void eval(char *cmdline)
{
    char *argv[MAXARGS]; /* Argument list execve() */
    char buf[MAXLINE];    /* Holds modified command line */
    int bg;               /* Should the job run in bg or fg? */
    pid_t pid;            /* Process id */

    strcpy(buf, cmdline);
    bg = parseline(buf, argv);
    if (argv[0] == NULL)
        return; /* Ignore empty lines */

    if (!builtin_command(argv)) {
        if ((pid = Fork()) == 0) { /* Child runs user job */
            if (execve(argv[0], argv, environ) < 0) {
                printf("%s: Command not found.\n", argv[0]);
                exit(0);
            }
        }

        /* Parent waits for foreground job to terminate */
        if (!bg) {
            int status;
            if (waitpid(pid, &status, 0) < 0)
                unix_error("waitfg: waitpid error");
        }
        else
            printf("%d %s", pid, cmdline);
    }
    return;
}
```

shellex.c

Simple Shell eval Function

```
void eval(char *cmdline)
{
    char *argv[MAXARGS]; /* Argument list execve() */
    char buf[MAXLINE];    /* Holds modified command line */
    int bg;               /* Should the job run in bg or fg? */
    pid_t pid;            /* Process id */

    strcpy(buf, cmdline);
    bg = parseline(buf, argv);
    if (argv[0] == NULL)
        return; /* Ignore empty lines */

    if (!builtin_command(argv)) {
        if ((pid = Fork()) == 0) { /* Child runs user job */
            if (execve(argv[0], argv, environ) < 0) {
                printf("%s: Command not found.\n", argv[0]);
                exit(0);
            }
        }

        /* Parent waits for foreground job to terminate */
        if (!bg) {
            int status;
            if (waitpid(pid, &status, 0) < 0)
                unix_error("waitfg: waitpid error");
        }
        else
            printf("%d %s", pid, cmdline);
    }
    return;
}
```

parseline will parse 'buf' into 'argv' and return whether or not input line ended in '&'

shellex.c

Simple Shell eval Function

```
void eval(char *cmdline)
{
    char *argv[MAXARGS]; /* Argument list execve() */
    char buf[MAXLINE];    /* Holds modified command line */
    int bg;               /* Should the job run in bg or fg? */
    pid_t pid;            /* Process id */

    strcpy(buf, cmdline);
    bg = parseline(buf, argv);
    if (argv[0] == NULL)
        return; /* Ignore empty lines */

    if (!builtin_command(argv)) {
        if ((pid = Fork()) == 0) { /* Child runs user job */
            if (execve(argv[0], argv, environ) < 0) {
                printf("%s: Command not found.\n", argv[0]);
                exit(0);
            }
        }

        /* Parent waits for foreground job to terminate */
        if (!bg) {
            int status;
            if (waitpid(pid, &status, 0) < 0)
                unix_error("waitfg: waitpid error");
        }
        else
            printf("%d %s", pid, cmdline);
    }
    return;
}
```

Ignore empty lines.

shellex.c

Simple Shell eval Function

```
void eval(char *cmdline)
{
    char *argv[MAXARGS]; /* Argument list execve() */
    char buf[MAXLINE];    /* Holds modified command line */
    int bg;               /* Should the job run in background */
    pid_t pid;            /* Process id */

    strcpy(buf, cmdline);
    bg = parseline(buf, argv);
    if (argv[0] == NULL)
        return; /* Ignore empty lines */

    if (!builtin_command(argv)) {
        if ((pid = Fork()) == 0) { /* Child runs user job */
            if (execve(argv[0], argv, environ) < 0) {
                printf("%s: Command not found.\n", argv[0]);
                exit(0);
            }
        }

        /* Parent waits for foreground job to terminate */
        if (!bg) {
            int status;
            if (waitpid(pid, &status, 0) < 0)
                unix_error("waitfg: waitpid error");
        }
        else
            printf("%d %s", pid, cmdline);
    }
    return;
}
```

If it is a 'built in' command, then handle it here in this program. Otherwise fork/exec the program specified in argv[0]

shellex.c

Simple Shell eval Function

```
void eval(char *cmdline)
{
    char *argv[MAXARGS]; /* Argument list execve() */
    char buf[MAXLINE];    /* Holds modified command line */
    int bg;               /* Should the job run in bg or fg? */
    pid_t pid;            /* Process id */

    strcpy(buf, cmdline);
    bg = parseline(buf, argv);
    if (argv[0] == NULL)
        return; /* Ignore empty lines */

    if (!builtin_command(argv)) {
        if ((pid = Fork()) == 0) { /* Child runs user job */
            if (execve(argv[0], argv, environ) < 0) {
                printf("%s: Command not found.\n", argv[0]);
                exit(0);
            }
        }

        /* Parent waits for foreground job to terminate */
        if (!bg) {
            int status;
            if (waitpid(pid, &status, 0) < 0)
                unix_error("waitfg: waitpid error");
        }
        else
            printf("%d %s", pid, cmdline);
    }
    return;
}
```

Create child

Simple Shell eval Function

```
void eval(char *cmdline)
{
    char *argv[MAXARGS]; /* Argument list execve() */
    char buf[MAXLINE];    /* Holds modified command line */
    int bg;               /* Should the job run in bg or fg? */
    pid_t pid;            /* Process id */

    strcpy(buf, cmdline);
    bg = parseline(buf, argv);
    if (argv[0] == NULL)
        return; /* Ignore empty lines */

    if (!builtin_command(argv)) {
        if ((pid = Fork()) == 0) { /* Child runs user job */
            if (execve(argv[0], argv, environ) < 0) {
                printf("%s: Command not found.\n", argv[0]);
                exit(0);
            }
        }

        /* Parent waits for foreground job to terminate */
        if (!bg) {
            int status;
            if (waitpid(pid, &status, 0) < 0)
                unix_error("waitfg: waitpid error");
        }
        else
            printf("%d %s", pid, cmdline);
    }
    return;
}
```

Start argv[0].
execve only returns on error.

Simple Shell eval Function

```
void eval(char *cmdline)
{
    char *argv[MAXARGS]; /* Argument list execve() */
    char buf[MAXLINE];    /* Holds modified command line */
    int bg;               /* Should the job run in bg or fg? */
    pid_t pid;            /* Process id */

    strcpy(buf, cmdline);
    bg = parseline(buf, argv);
    if (argv[0] == NULL)
        return; /* Ignore empty lines */

    if (!builtin_command(argv)) {
        if ((pid = Fork()) == 0) { /* Child runs user job */
            if (execve(argv[0], argv, environ) < 0) {
                printf("%s: Command not found.\n", argv[0]);
                exit(0);
            }
        }

        /* Parent waits for foreground job to terminate */
        if (!bg) {
            int status;
            if (waitpid(pid, &status, 0) < 0)
                unix_error("waitfg: waitpid error");
        }
        else
            printf("%d %s", pid, cmdline);
    }
    return;
}
```

If running child in foreground, wait until it is done.

shellex.c

Simple Shell eval Function

```
void eval(char *cmdline)
{
    char *argv[MAXARGS]; /* Argument list execve() */
    char buf[MAXLINE];    /* Holds modified command line */
    int bg;               /* Should the job run in bg or fg? */
    pid_t pid;            /* Process id */

    strcpy(buf, cmdline);
    bg = parseline(buf, argv);
    if (argv[0] == NULL)
        return; /* Ignore empty lines */

    if (!builtin_command(argv)) {
        if ((pid = Fork()) == 0) { /* Child runs user job */
            if (execve(argv[0], argv, environ) < 0) {
                printf("%s: Command not found.\n", argv[0]);
                exit(0);
            }
        }

        /* Parent waits for foreground job to terminate */
        if (!bg) {
            int status;
            if (waitpid(pid, &status, 0) < 0)
                unix_error("waitfg: waitpid error");
        }
        else
            printf("%d %s", pid, cmdline);
    }
    return;
}
```

If running child in background, print pid and continue doing other stuff.

shellex.c

Simple Shell eval Function

```
void eval(char *cmdline)
{
    char *argv[MAXARGS]; /* Argument list execve() */
    char buf[MAXLINE];    /* Holds modified command line */
    int bg;               /* Should the job run in bg or fg? */
    pid_t pid;            /* Process id */

    strcpy(buf, cmdline);
    bg = parseline(buf, argv);
    if (argv[0] == NULL)
        return; /* Ignore empty lines */

    if (!builtin_command(argv)) {
        if ((pid = Fork()) == 0) { /* Child runs user job */
            if (execve(argv[0], argv, environ) < 0) {
                printf("%s: Command not found.\n", argv[0]);
                exit(0);
            }
        }

        /* Parent waits for foreground job to terminate */
        if (!bg) {
            int status;
            if (waitpid(pid, &status, 0) < 0)
                unix_error("waitfg: waitpid error");
        }
        else
            printf("%d %s", pid, cmdline);
    }
    return;
}
```

Oops. There is a problem with this code.

shellex.c

Problem with Simple Shell Example

- **Shell designed to run indefinitely**
 - Should not accumulate unneeded resources
 - Memory
 - Child processes
 - File descriptors
- **Our example shell correctly waits for and reaps foreground jobs**
- **But what about background jobs?**
 - Will become zombies when they terminate
 - Will never be reaped because shell (typically) will not terminate
 - Will create a memory leak that could run the kernel out of memory

ECF to the Rescue!

■ Solution: Exceptional control flow

- The kernel will interrupt regular processing to alert us when a background process completes
- In Unix, the alert mechanism is called a *signal*

Today

- Shells
- **Signals**
- Nonlocal jumps

Signals

- A **signal** is a small message that notifies a process that an event of some type has occurred in the system
 - Akin to exceptions and interrupts
 - Sent from the kernel (sometimes at the request of another process) to a process
 - Signal type is identified by small integer ID's (1-30)
 - Only information in a signal is its ID and the fact that it arrived

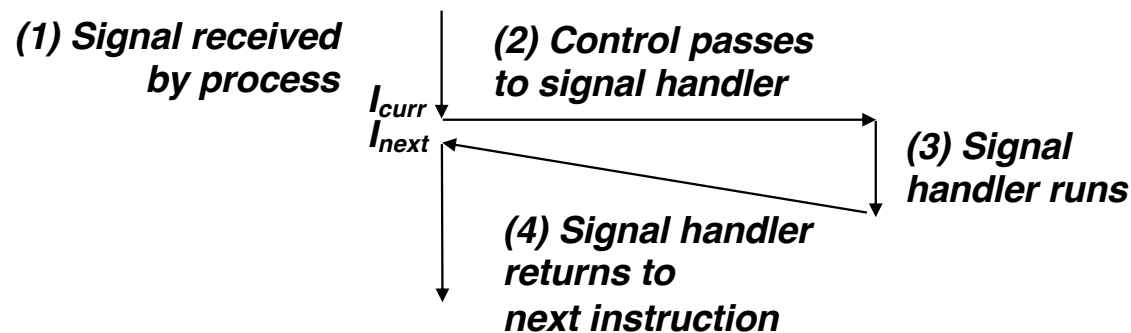
<i>ID</i>	<i>Name</i>	<i>Default Action</i>	<i>Corresponding Event</i>
2	SIGINT	Terminate	User typed ctrl-c
9	SIGKILL	Terminate	Kill program (cannot override or ignore)
11	SIGSEGV	Terminate	Segmentation violation
14	SIGALRM	Terminate	Timer signal
17	SIGCHLD	Ignore	Child stopped or terminated

Signal Concepts: Sending a Signal

- Kernel *sends* (delivers) a signal to a *destination process* by updating some state in the context of the destination process
- Kernel sends a signal for one of the following reasons:
 - Kernel has detected a system event such as divide-by-zero (SIGFPE) or the termination of a child process (SIGCHLD)
 - Another process has invoked the `kill` system call to explicitly request the kernel to send a signal to the destination process

Signal Concepts: Receiving a Signal

- A destination process *receives* a signal when it is forced by the kernel to react in some way to the delivery of the signal
- Some possible ways to react:
 - *Ignore* the signal (do nothing)
 - *Terminate* the process (with optional core dump)
 - *Catch* the signal by executing a user-level function called *signal handler*
 - Akin to a hardware exception handler being called in response to an asynchronous interrupt:



Signal Concepts: Pending and Blocked Signals

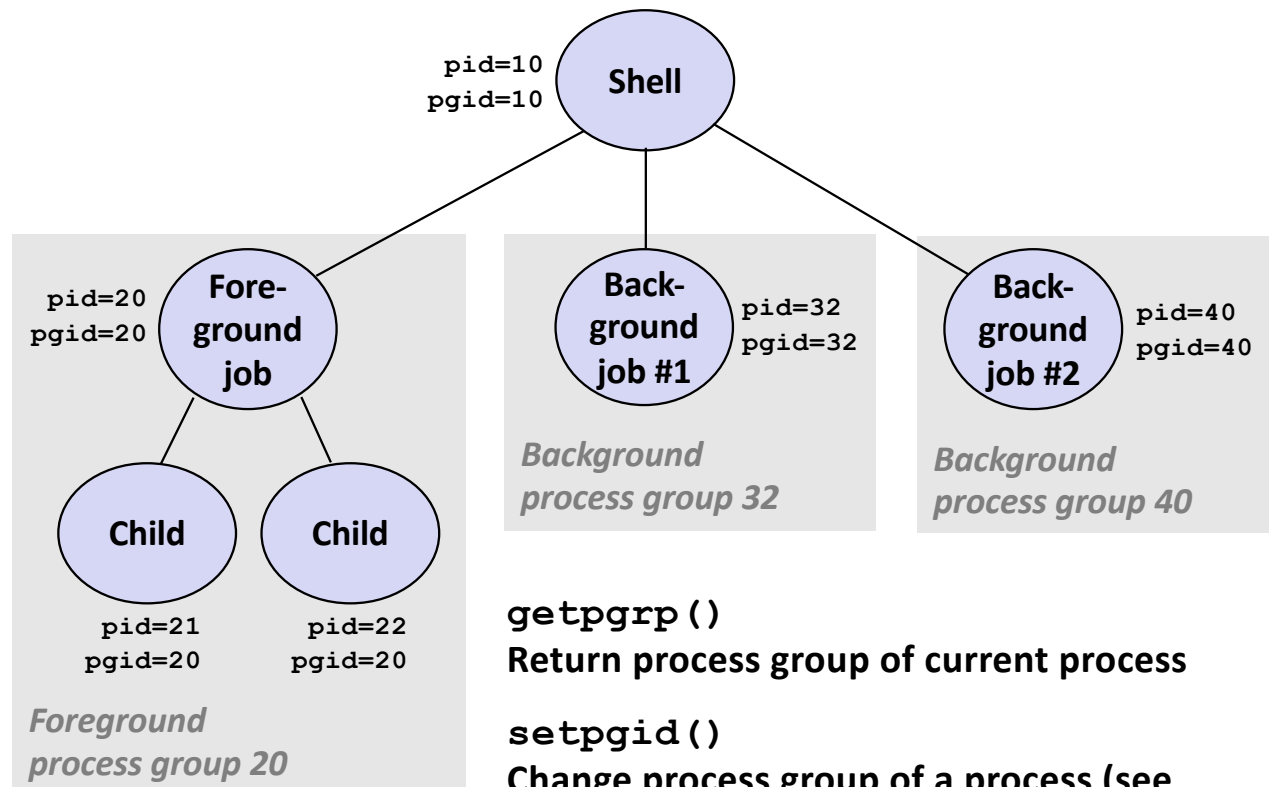
- A signal is **pending** if sent but not yet received
 - There can be at most one pending signal of any particular type
 - Important: Signals are not queued
 - If a process has a pending signal of type k, then subsequent signals of type k that are sent to that process are discarded
- A process can **block** the receipt of certain signals
 - Blocked signals can be delivered, but will not be received until the signal is unblocked
- A pending signal is received at most once

Signal Concepts: Pending/Blocked Bits

- Kernel maintains **pending** and **blocked** bit vectors in the context of each process
 - **pending**: represents the set of pending signals
 - Kernel sets bit *k* in **pending** when a signal of type *k* is delivered
 - Kernel clears bit *k* in **pending** when a signal of type *k* is received
 - **blocked**: represents the set of blocked signals
 - Can be set and cleared by using the **sigprocmask** function
 - Also referred to as the *signal mask*.

Sending Signals: Process Groups

- Every process belongs to exactly one process group



Sending Signals with `/bin/kill` Program

- `/bin/kill` program
sends arbitrary signal to a
process or process group

- **Examples**

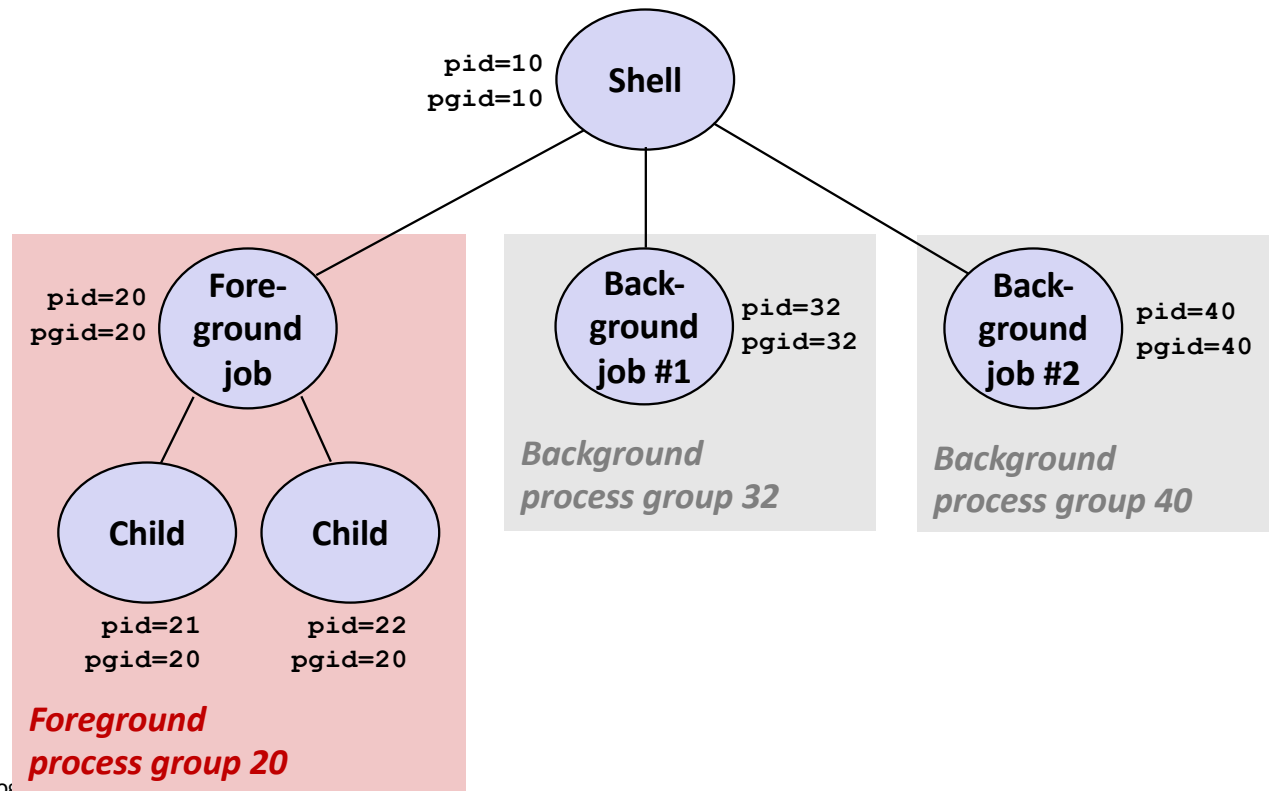
- `/bin/kill -9 24818`
Send SIGKILL to process 24818
- `/bin/kill -9 -24817`
Send SIGKILL to every process
in process group 24817

```
linux> ./forks 16
Child1: pid=24818 pgrp=24817
Child2: pid=24819 pgrp=24817

linux> ps
  PID TTY          TIME CMD
 24788 pts/2        00:00:00 tcsh
 24818 pts/2        00:00:02 forks
 24819 pts/2        00:00:02 forks
 24820 pts/2        00:00:00 ps
linux> /bin/kill -9 -24817
linux> ps
  PID TTY          TIME CMD
 24788 pts/2        00:00:00 tcsh
 24823 pts/2        00:00:00 ps
linux>
```

Sending Signals from the Keyboard

- Typing ctrl-c (ctrl-z) causes the kernel to send a SIGINT (SIGTSTP) to every job in the foreground process group.
 - SIGINT – default action is to terminate each process
 - SIGTSTP – default action is to stop (suspend) each process



Example of `ctrl-c` and `ctrl-z`

```
bluefish> ./forks 17
Child: pid=28108 pgrp=28107
Parent: pid=28107 pgrp=28107
<types ctrl-z>
Suspended
bluefish> ps w
  PID TTY          STAT TIME  COMMAND
 27699 pts/8        Ss   0:00  -tcsh
 28107 pts/8        T    0:01  ./forks 17
 28108 pts/8        T    0:01  ./forks 17
 28109 pts/8        R+   0:00  ps w
bluefish> fg
./forks 17
<types ctrl-c>
bluefish> ps w
  PID TTY          STAT TIME  COMMAND
 27699 pts/8        Ss   0:00  -tcsh
 28110 pts/8        R+   0:00  ps w
```

STAT (process state) Legend:

First letter:

S: sleeping

T: stopped

R: running

Second letter:

s: session leader

+: foreground proc group

See “man ps” for more details

Sending Signals with `kill` Function

```
void fork12()
{
    pid_t pid[N];
    int i;
    int child_status;

    for (i = 0; i < N; i++)
        if ((pid[i] = fork()) == 0) {
            /* Child: Infinite Loop */
            while(1)
                ;
        }

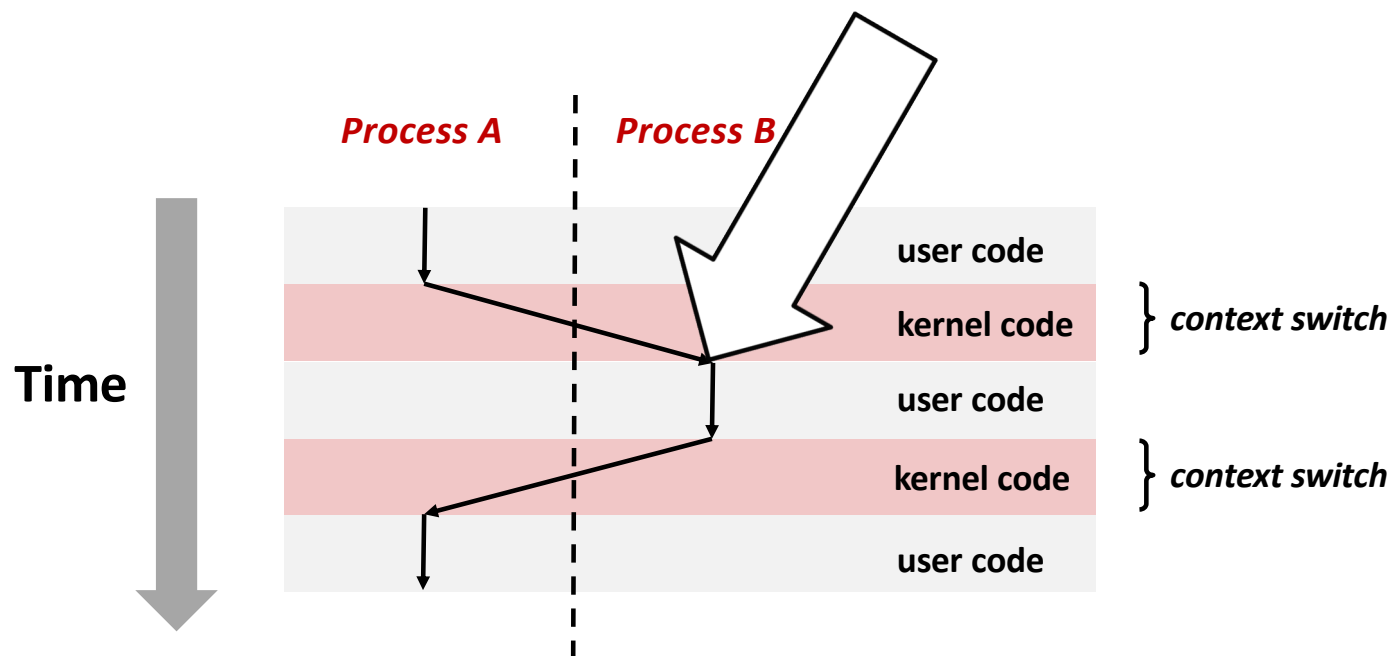
    for (i = 0; i < N; i++) {
        printf("Killing process %d\n", pid[i]);
        kill(pid[i], SIGINT);
    }

    for (i = 0; i < N; i++) {
        pid_t wpid = wait(&child_status);
        if (WIFEXITED(child_status))
            printf("Child %d terminated with exit status %d\n",
                wpid, WEXITSTATUS(child_status));
        else
            printf("Child %d terminated abnormally\n", wpid);
    }
}
```

forks.c

Receiving Signals

- Suppose kernel is returning from an exception handler and is ready to pass control to process p



Receiving Signals

- Suppose kernel is returning from an exception handler and is ready to pass control to process p
- Kernel computes $\text{pnb} = \text{pending} \ \& \ \sim\text{blocked}$
 - The set of pending nonblocked signals for process p
- If ($\text{pnb} == 0$)
 - Pass control to next instruction in the logical flow for p
- Else
 - Choose least nonzero bit k in pnb and force process p to *receive* signal k
 - The receipt of the signal triggers some *action* by p
 - Repeat for all nonzero k in pnb
 - Pass control to next instruction in logical flow for p

Default Actions

- Each signal type has a predefined *default action*, which is one of:
 - The process terminates
 - The process stops until restarted by a SIGCONT signal
 - The process ignores the signal

Installing Signal Handlers

- The `signal` function modifies the default action associated with the receipt of signal `signum`:
 - `handler_t *signal(int signum, handler_t *handler)`
- Different values for `handler`:
 - `SIG_IGN`: ignore signals of type `signum`
 - `SIG_DFL`: revert to the default action on receipt of signals of type `signum`
 - Otherwise, `handler` is the address of a user-level *signal handler*
 - Called when process receives signal of type `signum`
 - Referred to as *“installing”* the handler
 - Executing handler is called *“catching”* or *“handling”* the signal
 - When the handler executes its return statement, control passes back to instruction in the control flow of the process that was interrupted by receipt of the signal

Signal Handling Example

```
void sigint_handler(int sig) /* SIGINT handler */
{
    printf("So you think you can stop the bomb with ctrl-c, do  
you?\n");
    sleep(2);
    printf("Well...\n");
    fflush(stdout);
    sleep(1);
    printf("OK. :-)\n");
    exit(0);
}

int main()
{
    /* Install the SIGINT handler */
    if (signal(SIGINT, sigint_handler) == SIG_ERR)
        unix_error("signal error");

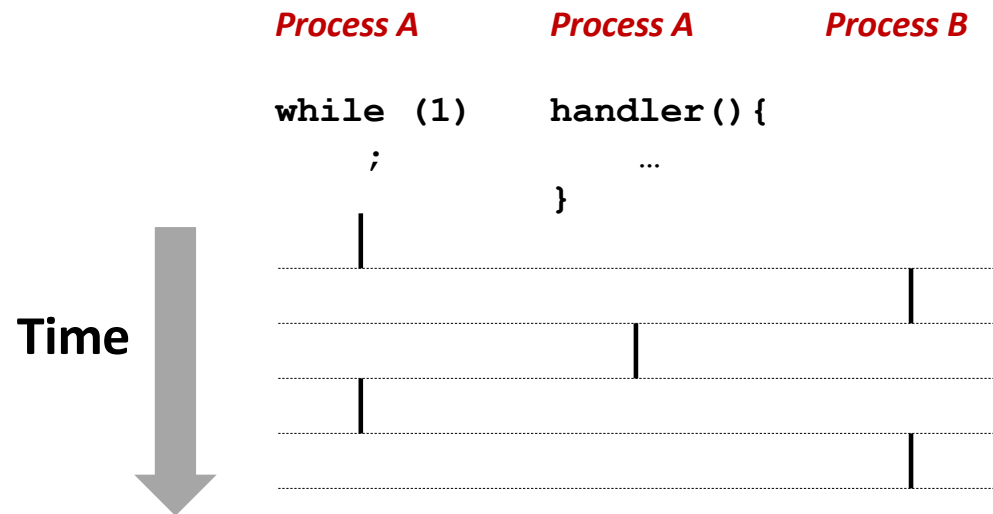
    /* Wait for the receipt of a signal */
    pause();

    return 0;
}
```

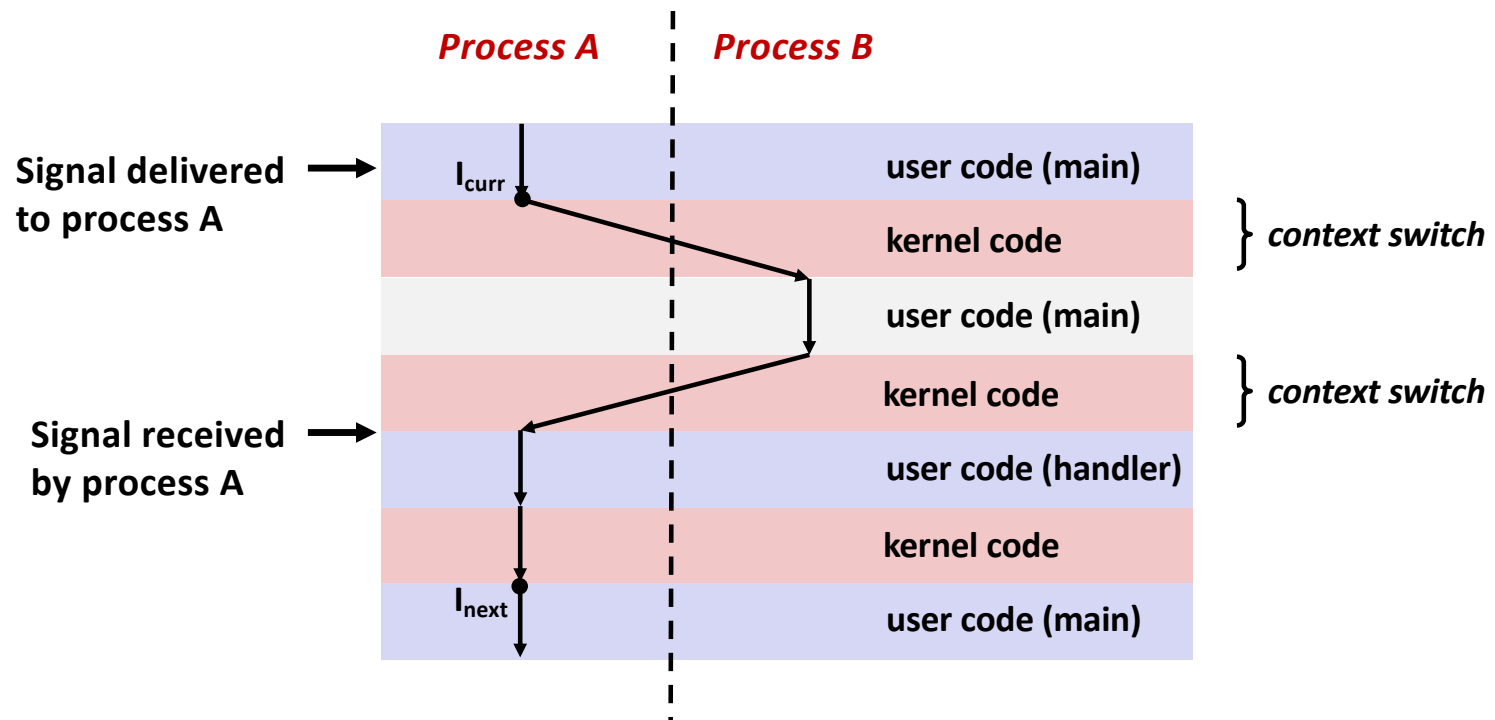
sigint.c

Signals Handlers as Concurrent Flows

- A signal handler is a separate logical flow (not process) that runs concurrently with the main program

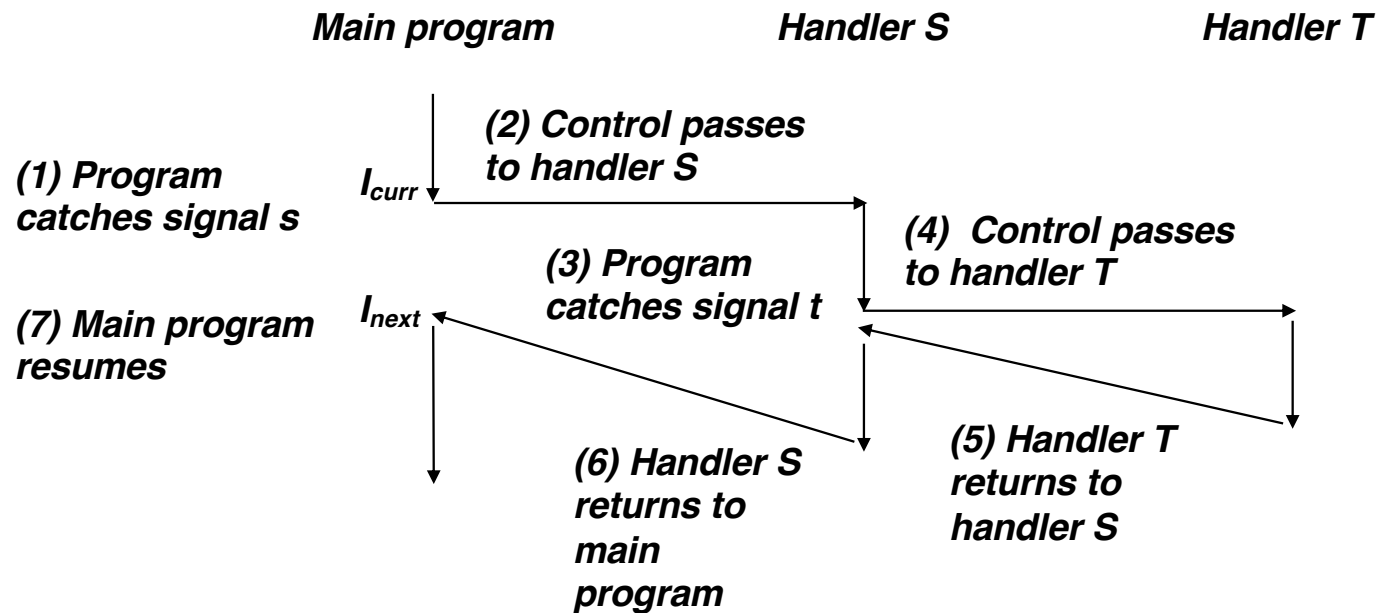


Another View of Signal Handlers as Concurrent Flows



Nested Signal Handlers

- Handlers can be interrupted by other handlers



Blocking and Unblocking Signals

■ Implicit blocking mechanism

- Kernel blocks any pending signals of type currently being handled.
- E.g., A SIGINT handler can't be interrupted by another SIGINT

■ Explicit blocking and unblocking mechanism

- `sigprocmask` function

■ Supporting functions

- `sigemptyset` – Create empty set
- `sigfillset` – Add every signal number to set
- `sigaddset` – Add signal number to set
- `sigdelset` – Delete signal number from set

Temporarily Blocking Signals

```
sigset_t mask, prev_mask;

Sigemptyset(&mask);
Sigaddset(&mask, SIGINT);

/* Block SIGINT and save previous blocked set */
Sigprocmask(SIG_BLOCK, &mask, &prev_mask);

:   /* Code region that will not be interrupted by SIGINT
*/

/* Restore previous blocked set, unblocking SIGINT */
Sigprocmask(SIG_SETMASK, &prev_mask, NULL);
```

Safe Signal Handling

- **Handlers are tricky because they are concurrent with main program and share the same global data structures.**
 - Shared data structures can become corrupted.
- **For now here are some guidelines to help you avoid trouble.**

Guidelines for Writing Safe Handlers

- **G0: Keep your handlers as simple as possible**
 - e.g., Set a global flag and return
- **G1: Call only async-signal-safe functions in your handlers**
 - `printf`, `sprintf`, `malloc`, and `exit` are not safe!
- **G2: Save and restore `errno` on entry and exit**
 - So that other handlers don't overwrite your value of `errno`
- **G3: Protect accesses to shared data structures by temporarily blocking all signals.**
 - To prevent possible corruption
- **G4: Declare global variables as `volatile`**
 - To prevent compiler from storing them in a register
- **G5: Declare global flags as `volatile sig_atomic_t`**
 - *flag*: variable that is only read or written (e.g. `flag = 1`, not `flag++`)
 - Flag declared this way does not need to be protected like other globals

Async-Signal-Safety

- Function is *async-signal-safe* if either reentrant (e.g., all variables stored on stack frame, CS:APP3e 12.7.2) or non-interruptible by signals.
- Posix guarantees 117 functions to be async-signal-safe
 - Source: “man 7 signal”
 - Popular functions on the list:
 - `_exit`, `write`, `wait`, `waitpid`, `sleep`, `kill`
 - Popular functions that are **not** on the list:
 - `printf`, `sprintf`, `malloc`, `exit`
 - Unfortunate fact: `write` is the only async-signal-safe output function

Safely Generating Formatted Output

- Use the reentrant SIO (Safe I/O library) from `csapp.c` in your handlers.

- `ssize_t sio_puts(char s[]) /* Put string */`
- `ssize_t sio_putl(long v) /* Put long */`
- `void sio_error(char s[]) /* Put msg & exit */`

```
void sigint_handler(int sig) /* Safe SIGINT handler */
{
    Sio_puts("So you think you can stop the bomb with
ctrl-c, do you?\n");
    sleep(2);
    Sio_puts("Well...\n");
    sleep(1);
    Sio_puts("OK. :-)\n");
    _exit(0);
}
```

sigintsafe.c

Correct Signal Handling

```
int ccount = 0;
void child_handler(int sig) {
    int olderrno = errno;
    pid_t pid;
    if ((pid = wait(NULL)) < 0)
        Sio_error("wait error");
    ccount--;
    Sio_puts("Handler reaped child ");
    Sio_putl((long)pid);
    Sio_puts(" \n");
    sleep(1);
    errno = olderrno;
}

void fork14() {
    pid_t pid[N];
    int i;
    ccount = N;
    Signal(SIGCHLD, child_handler);

    for (i = 0; i < N; i++) {
        if ((pid[i] = Fork()) == 0) {
            Sleep(1);
            exit(0); /* Child exits */
        }
    }
    while (ccount > 0) /* Parent spins */
        ;
}
```

forks.c

- Pending signals are not queued
 - For each signal type, one bit indicates whether or not signal is pending...
 - ...thus at most one pending signal of any particular type.
- You can't use signals to count events, such as children terminating.

```
whaleshark> ./forks 14
Handler reaped child 23240
Handler reaped child 23241
```

Correct Signal Handling

- **Must wait for all terminated child processes**

- Put `wait` in a loop to reap all terminated children

```
void child_handler2(int sig)
{
    int olderrno = errno;
    pid_t pid;
    while ((pid = wait(NULL)) > 0) {
        ccount--;
        Sio_puts("Handler reaped child ");
        Sio_putl((long)pid);
        Sio_puts(" \n");
    }
    if (errno != ECHILD)
        Sio_error("wait error");
    errno = olderrno;
}
```

```
whaleshark> ./forks 15
Handler reaped child 23246
Handler reaped child 23247
Handler reaped child 23248
Handler reaped child 23249
Handler reaped child 23250
whaleshark>
```

Portable Signal Handling

- **Ugh! Different versions of Unix can have different signal handling semantics**
 - Some older systems restore action to default after catching signal
 - Some interrupted system calls can return with `errno == EINTR`
 - Some systems don't block signals of the type being handled
- **Solution: `sigaction`**

```
handler_t *Signal(int signum, handler_t *handler)
{
    struct sigaction action, old_action;

    action.sa_handler = handler;
    sigemptyset(&action.sa_mask); /* Block sigs of type being handled */
    action.sa_flags = SA_RESTART; /* Restart syscalls if possible */

    if (sigaction(signum, &action, &old_action) < 0)
        unix_error("Signal error");
    return (old_action.sa_handler);
}
```

csapp.c

Synchronizing Flows to Avoid Races

- Simple shell with a subtle synchronization error because it assumes parent runs before child.

```
int main(int argc, char **argv)
{
    int pid;
    sigset_t mask_all, prev_all;

    Sigfillset(&mask_all);
    Signal(SIGCHLD, handler);
    initjobs(); /* Initialize the job list */

    while (1) {
        if ((pid = Fork()) == 0) { /* Child */
            Execve("/bin/date", argv, NULL);
        }
        Sigprocmask(SIG_BLOCK, &mask_all, &prev_all); /* Parent */
        addjob(pid); /* Add the child to the job list */
        Sigprocmask(SIG_SETMASK, &prev_all, NULL);
    }
    exit(0);
}
```

procmask1.c

Synchronizing Flows to Avoid Races

■ SIGCHLD handler for a simple shell

```
void handler(int sig)
{
    int olderrno = errno;
    sigset_t mask_all, prev_all;
    pid_t pid;

    Sigfillset(&mask_all);
    while ((pid = waitpid(-1, NULL, 0)) > 0) { /* Reap child */
        Sigprocmask(SIG_BLOCK, &mask_all, &prev_all);
        deletejob(pid); /* Delete the child from the job list */
        Sigprocmask(SIG_SETMASK, &prev_all, NULL);
    }
    if (errno != ECHILD)
        Sio_error("waitpid error");
    errno = olderrno;
}
```

procmask1.c

Corrected Shell Program without Race

```
int main(int argc, char **argv)
{
    int pid;
    sigset_t mask_all, mask_one, prev_one;

    Sigfillset(&mask_all);
    Sigemptyset(&mask_one);
    Sigaddset(&mask_one, SIGCHLD);
    Signal(SIGCHLD, handler);
    initjobs(); /* Initialize the job list */

    while (1) {
        Sigprocmask(SIG_BLOCK, &mask_one, &prev_one); /* Block SIGCHLD */
        if ((pid = Fork()) == 0) { /* Child process */
            Sigprocmask(SIG_SETMASK, &prev_one, NULL); /* Unblock SIGCHLD */
            Execve("/bin/date", argv, NULL);
        }
        Sigprocmask(SIG_BLOCK, &mask_all, NULL); /* Parent process */
        addjob(pid); /* Add the child to the job list */
        Sigprocmask(SIG_SETMASK, &prev_one, NULL); /* Unblock SIGCHLD */
    }
    exit(0);
}
```

procmask2.c

Explicitly Waiting for Signals

- Handlers for program explicitly waiting for SIGCHLD to arrive.

```
volatile sig_atomic_t pid;

void sigchld_handler(int s)
{
    int olderrno = errno;
    pid = Waitpid(-1, NULL, 0); /* Main is waiting for nonzero pid
*/
    errno = olderrno;
}

void sigint_handler(int s)
{
}
```

waitforsignal.c

Explicitly Waiting for Signals

```
int main(int argc, char **argv) {
    sigset_t mask, prev;
    Signal(SIGCHLD, sigchld_handler);
    Signal(SIGINT, sigint_handler);
    Sigemptyset(&mask);
    Sigaddset(&mask, SIGCHLD);

    while (1) {
        Sigprocmask(SIG_BLOCK, &mask, &prev); /* Block SIGCHLD */
        if (Fork() == 0) /* Child */
            exit(0);
        /* Parent */
        pid = 0;
        Sigprocmask(SIG_SETMASK, &prev, NULL); /* Unblock SIGCHLD */

        /* Wait for SIGCHLD to be received (wasteful!) */
        while (!pid)
            ;
        /* Do some work after receiving SIGCHLD */
        printf(".");
    }
    exit(0);
}
```

Similar to a shell waiting for a foreground job to terminate.

waitforsignal.c

Explicitly Waiting for Signals

- **Program is correct, but very wasteful**

- Program in busy-wait loop

```
while (!pid)
    ;
```

- **Possible race condition**

- Between checking pid and starting pause, might receive signal

```
while (!pid) /* Race! */
    pause();
```

- **Safe, but slow**

- Will take up to one second to respond

```
while (!pid) /* Too slow! */
    sleep(1);
```

Waiting for Signals with `sigsuspend`

- `int sigsuspend(const sigset_t *mask)`
- Equivalent to atomic (uninterruptable) version of:

```
sigprocmask(SIG_BLOCK, &mask, &prev);  
pause();  
sigprocmask(SIG_SETMASK, &prev, NULL);
```

Waiting for Signals with sigsuspend

```
int main(int argc, char **argv) {
    sigset_t mask, prev;
    Signal(SIGCHLD, sigchld_handler);
    Signal(SIGINT, sigint_handler);
    Sigemptyset(&mask);
    Sigaddset(&mask, SIGCHLD);

    while (1) {
        Sigprocmask(SIG_BLOCK, &mask, &prev); /* Block SIGCHLD */
        if (Fork() == 0) /* Child */
            exit(0);

        /* Wait for SIGCHLD to be received */
        pid = 0;
        while (!pid)
            Sigsuspend(&prev);

        /* Optionally unblock SIGCHLD */
        Sigprocmask(SIG_SETMASK, &prev, NULL);
        /* Do some work after receiving SIGCHLD */
        printf(".");
    }
    exit(0);
}
```

sigsuspend.c

Today

- Shells
- Signals
- **Nonlocal jumps**

Nonlocal Jumps: `setjmp/longjmp`

- **Powerful (but dangerous) user-level mechanism for transferring control to an arbitrary location**
 - Controlled to way to break the procedure call / return discipline
 - Useful for error recovery and signal handling
- **`int setjmp(jmp_buf j)`**
 - Must be called before `longjmp`
 - Identifies a return site for a subsequent `longjmp`
 - Called **once**, returns **one or more** times
- **Implementation:**
 - Remember where you are by storing the current **register context**, **stack pointer**, and **PC value** in `jmp_buf`
 - Return 0

setjmp/longjmp (cont)

■ `void longjmp(jmp_buf j, int i)`

- Meaning:
 - return from the `setjmp` remembered by jump buffer `j` again ...
 - ... this time returning `i` instead of 0
- Called after `setjmp`
- Called **once**, but **never** returns

■ `longjmp` Implementation:

- Restore register context (stack pointer, base pointer, PC value) from jump buffer `j`
- Set `%eax` (the return value) to `i`
- Jump to the location indicated by the PC stored in jump buf `j`

setjmp/longjmp Example

- Goal: return directly to original caller from a deeply-nested function

```
/* Deeply nested function foo
*/
void foo(void)
{
    if (error1)
        longjmp(buf, 1);
    bar();
}

void bar(void)
{
    if (error2)
        longjmp(buf, 2);
}
```

setjmp/longjmp Example (cont)

```
jmp_buf buf;

int error1 = 0;
int error2 = 1;

void foo(void), bar(void);

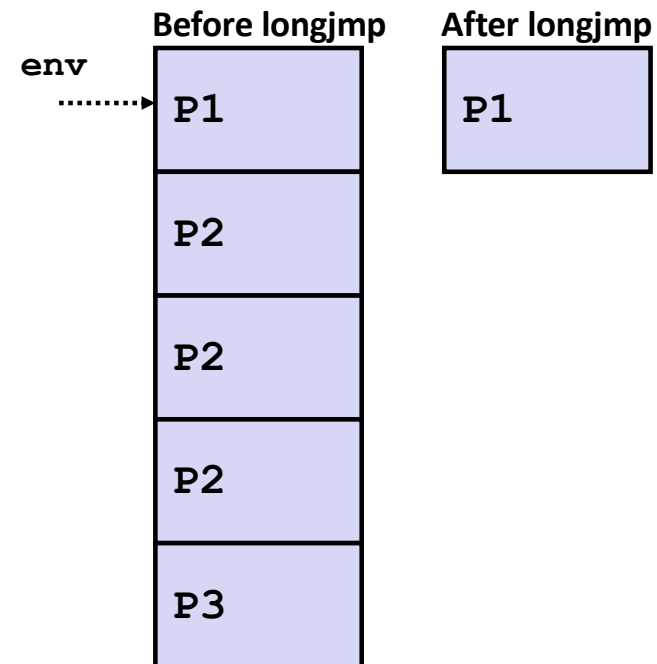
int main()
{
    switch(setjmp(buf)) {
        case 0:
            foo();
            break;
        case 1:
            printf("Detected an error1 condition in
foo\n");
            break;
        case 2:
            printf("Detected an error2 condition in
bar\n");
            break;
        default:
            printf("Unknown error condition in foo\n");
    }
    exit(0);
}
```

Limitations of Nonlocal Jumps

■ Works within stack discipline

- Can only long jump to environment of function that has been called but not yet completed

```
jmp_buf env;  
  
P1()  
{  
    if (setjmp(env)) {  
        /* Long Jump to here */  
    } else {  
        P2();  
    }  
}  
  
P2()  
{  
    . . . P2(); . . . P3();  
}  
  
P3()  
{  
    longjmp(env, 1);  
}
```

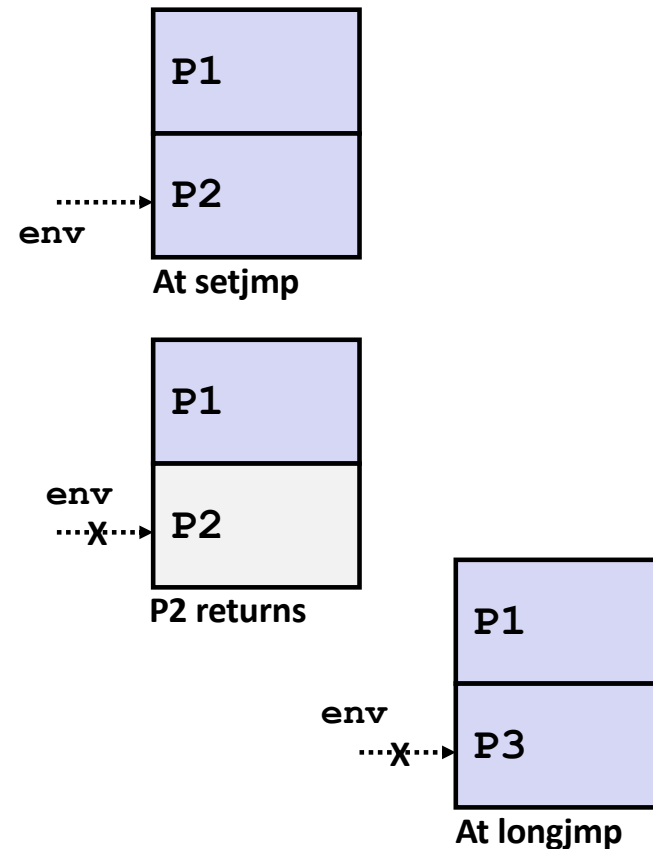


Limitations of Long Jumps (cont.)

■ Works within stack discipline

- Can only long jump to environment of function that has been called but not yet completed

```
jmp_buf env;  
  
P1()  
{  
    P2(); P3();  
}  
  
P2()  
{  
    if (setjmp(env)) {  
        /* Long Jump to here */  
    }  
}  
  
P3()  
{  
    longjmp(env, 1);  
}
```



Putting It All Together: A Program That Restarts Itself When `ctrl-c`'d

```
greatwhite> ./restart
starting
processing...
processing...
processing...
restarting ← Ctrl-c
processing...
processing...
restarting ← Ctrl-c
processing...
processing...
processing...
```

`restart.c`

Putting It All Together: A Program That Restarts Itself When `ctrl-c`'d

```
#include "csapp.h"

sigjmp_buf buf;

void handler(int sig)
{
    siglongjmp(buf, 1);
}

int main()
{
    if (!sigsetjmp(buf, 1)) {
        Signal(SIGINT, handler);
        Sio_puts("starting\n");
    }
    else
        Sio_puts("restarting\n");

    while(1) {
        Sleep(1);
        Sio_puts("processing...\n");
    }
    exit(0); /* Control never reaches here */
}

restart.c
```

Summary

- **Signals provide process-level exception handling**
 - Can generate from user programs
 - Can define effect by declaring signal handler
 - Be very careful when writing signal handlers
- **Nonlocal jumps provide exceptional control flow within process**
 - Within constraints of stack discipline